

PhysFormer: A Physics-Guided Transformer for Physically Consistent Virtual Power Plant Forecasting

Xingyu Liu^{a,*}, Zhong Tang^a

^aShanghai University of Electric Power, Department of Electrical Engineering, Shanghai, 200090, China

Abstract

Virtual power plant (VPP) forecasting must jointly predict load, photovoltaic (PV), and wind power while avoiding non-physical outputs such as negative generation or nighttime solar power. Simple post-processing can remove some violations at deployment, but it does not constrain the training dynamics or clarify how weather priors influence model behavior. This paper presents PhysFormer, a physics-guided Transformer for physically consistent VPP forecasting. The framework combines an explicit physical mapping stream, a bounded residual output mechanism, a physics-guided gating and alignment module, and curriculum-based constraint training. The bounded residual design keeps predictions non-negative under the adopted normalization setting, while the gating and regularization terms align PV and wind activation with exogenous weather cues. Experiments on a real-world 3-year VPP dataset show that PhysFormer achieves a boundary violation rate of 0.00% and remains close in MSE to strong Transformer baselines, with an

*Corresponding author.

Email addresses: y24203228@mail.shiep.edu.cn (Xingyu Liu), tangzhong64@163.com (Zhong Tang)

MSE of 0.0128 compared with 0.0121 and 0.0122 for Informer and iTransformer, respectively. These results indicate that the proposed framework provides a practical trade-off among forecasting accuracy, physical consistency, and interpretable weather-response behavior for joint VPP component prediction.

Keywords: Virtual power plant, Forecasting, Physics-guided learning, Physical consistency, Transformer, Renewable power

1. Introduction

1.1. Background and Motivation

The integration of renewable energy sources into the power grid has driven the development of Virtual Power Plants (VPPs) to aggregate and manage distributed energy resources. Accurate forecasting of VPP dynamics—specifically electrical load, photovoltaic (PV) generation, and wind power generation—is critical for grid stability, economic dispatch, and operational risk management [1, 2, 3, 4, 5]. Unlike single-variable forecasting, cooperative VPP management requires simultaneous, coherent predictions across these interrelated domains, where the physical constraints of each energy modality must be rigorously respected. For instance, PV generation is intrinsically bound by solar irradiance and cannot produce power during nighttime, while wind generation depends on a non-linear turbine power curve driven by localized wind speed.

Historically, early approaches to VPP forecasting relied on statistical extrapolation. As distributed resources increased, recurrent neural networks and Transformer-based frameworks became widely used for sequential mod-

eling, especially when long-range temporal dependencies and multi-source interactions must be captured [6, 7, 8, 9]. Recent application-oriented studies have continued to adapt these architectures to load forecasting and VPP scheduling scenarios [3, 10, 11, 12]. In these settings, low statistical error is important, but so is maintaining predictions that remain consistent with basic physical operating conditions.

1.2. Challenges of Data-Driven VPP Forecasting

Recent advances in deep learning have improved time-series forecasting accuracy, yet their application to VPP prediction still involves several practical challenges [13, 14]:

1) Output feasibility and activation consistency: Purely data-driven predictors can achieve competitive error metrics while still producing physically inconsistent outputs, such as non-zero solar generation at night or negative renewable power. In VPP applications, these violations may be small in magnitude, but they complicate downstream interpretation and operational use.

2) Limits of post-processing and static penalties: Post-hoc clipping is a practical remedy for deployment, but it does not guide the model during training. Conversely, physics-informed penalties can regularize optimization, yet fixed constraints may not be sufficient to reflect weather-driven activation changes in PV and wind generation [15, 16, 17]. A useful forecasting architecture should therefore balance train-time guidance with flexible residual modeling.

3) Need for interpretable weather-response behavior: For deployment-oriented forecasting, it is helpful to understand whether activation patterns

in the model remain aligned with observable weather cues. Existing interpretability tools are informative, but they do not necessarily encode these relationships into the model structure itself [13, 18, 14].

1.3. Overview of This Work

To address these issues, this paper presents **PhysFormer**, a physics-guided Transformer for physically consistent VPP forecasting. The proposed framework combines an explicit physical mapping stream, a bounded residual output parameterization, a physics-guided gating and alignment module (PGCC), and curriculum-based constraint training. In this work, PGCC is used as a guidance and alignment mechanism rather than as a causal discovery tool.

The main contributions of this paper are as follows:

1. We propose a **Bounded Physical Act-Residual (BPAR)** output mechanism that combines theory-based baselines with data-driven residual corrections, reducing non-physical predictions while preserving differentiable training dynamics.
2. We introduce a **Physics-Guided Gating and Alignment** module (PGCC) that injects weather-dependent priors into PV and wind activation, improving the interpretability of the model’s weather-response behavior through supervised alignment.

In addition, we adopt a **Physics-Constrained Curriculum Training** strategy to stabilize multi-objective optimization. Experiments on a real 3-year VPP dataset show that PhysFormer reaches zero boundary violations while remaining close in MSE to strong Transformer baselines, illustrating

a practical trade-off between forecasting accuracy, physical consistency, and engineering interpretability.

2. Proposed PhysFormer Framework

2.1. Mathematical Formulation

Throughout this paper, we adopt the following notation: P_{load} , P_{pv} , and P_{wind} denote load, PV, and wind power [MW]; G , T , and v represent irradiance [W/m^2], temperature [$^{\circ}\text{C}$], and wind speed [m/s]; η_{pv} and β_T are the PV conversion efficiency and temperature coefficient; v_{ci} , v_r , and v_{co} are the wind turbine cut-in, rated, and cut-out speeds; a_{pv} and a_{wind} are physics activity masks $\in [0, 1]$; g_{pv} and g_{wind} are PGCC gates; α is the curriculum factor; $S = 672$ and $P = 96$ are the sequence and prediction lengths; and $D = 512$ is the model dimension.

In the context of Virtual Power Plant (VPP) forecasting, the objective is to predict the future power outputs of three distinct sub-components: base load (P_{load}), photovoltaic generation (P_{pv}), and wind generation (P_{wind}). The overall net power demand of the VPP is defined as $P_{\text{net}} = P_{\text{load}} - P_{\text{pv}} - P_{\text{wind}}$. Let $\mathcal{X}_{\text{stat}} \in \mathbb{R}^{B \times S \times 3}$ denote the historical observation sequence for the three components over a look-back window of length S , where B is the batch size. Concurrently, the corresponding historical weather conditions—comprising temperature, irradiance, and wind speed—are represented by $\mathcal{X}_{\text{weather_hist}} \in \mathbb{R}^{B \times S \times 3}$. The model is also provided with future weather forecasts for the prediction horizon P , denoted by $\mathcal{X}_{\text{weather_future}} \in \mathbb{R}^{B \times P \times 3}$. Given these inputs, the forecasting model aims to generate point predictions for the output $Y \in \mathbb{R}^{B \times P \times 3}$, providing the respective future trajectories for the load, PV,

and wind components.

2.2. Overall Architecture

The PhysFormer is constructed as a dual-stream Transformer with PGCC and theory-anchored activity gating. The processing pipeline follows seven steps: (1) Historical observation $\mathcal{X}_{\text{stat}}$ and weather $\mathcal{X}_{\text{weather_hist}}$ are concatenated and encoded by a Transformer encoder ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, 8 heads, 3 layers). (2) $\mathcal{X}_{\text{weather_hist}}$ is processed through ExplicitPhysicalMapping for physical baselines. (3) PGCC fuses statistical and physics representations. (4) FlattenHead projects from historical dimension S to prediction horizon P . (5) Future weather $\mathcal{X}_{\text{weather_future}}$ is injected via a GLU with residual logic. (6) Separate output heads predict per-component residual corrections. (7) Theory-Anchored Output applies physics activity masks to yield final predictions.

2.3. Explicit Physical Mapping

The core of the physics injection lies in the ExplicitPhysicalMapping layer, which maps relevant weather variables to theoretical power baselines via domain-specific equations.

2.3.1. PV Model

The theoretical PV generation is driven by irradiance G and degraded by cell temperature, modeled as:

$$P_{\text{pv}} = G \cdot \eta_{\text{pv}} \cdot \text{clamp}(1 - \beta_T \cdot (T - 25), 0, 1.5) \quad (1)$$

where T is the temperature, $\eta_{\text{pv}} = 0.33 \cdot \sigma(\theta_{\text{eff}})$ is the learnable conversion efficiency bounded by the Shockley-Queisser limit, and $\beta_T = \text{softplus}(\theta_{\text{temp_coef}}) \times$

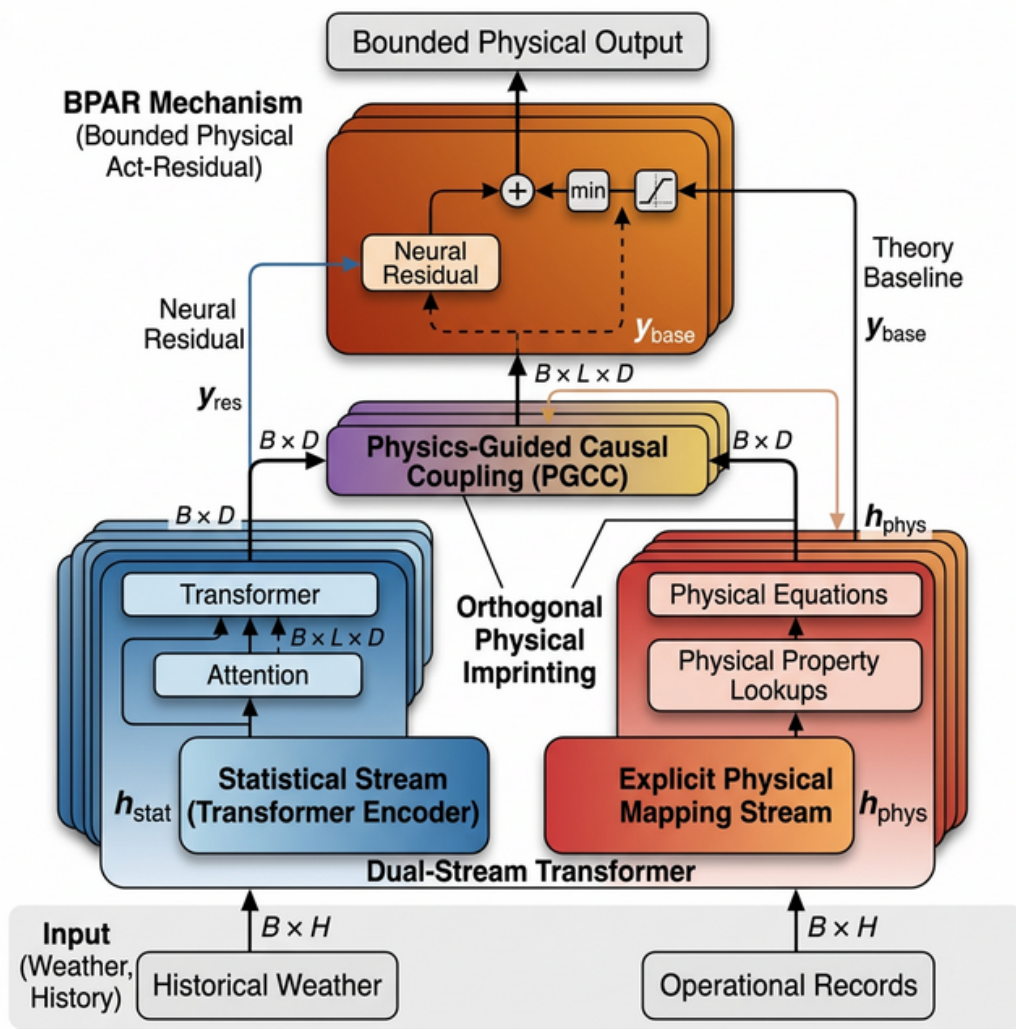


Figure 1: The overall architecture of the proposed PhysFormer. It features a dual-stream design: a statistical stream encoded by a Transformer and an explicit physical mapping stream. These streams are fused through the Physics-Guided Gating and Alignment (PGCC) module. Future weather conditions are injected using a Gated Linear Unit (GLU) to provide explicit foresight. Finally, theory-anchored residual heads refine the predictions under physics-derived activity gates to reduce non-physical outputs.

$0.01 \in [0.003, 0.005]/^\circ\text{C}$ is the temperature coefficient.

2.3.2. Wind Model

Wind power limits depend heavily on operating thresholds: the cut-in speed v_{ci} , rated speed v_{r} , and cut-out speed v_{co} . To preserve strict threshold ordering, we utilize a cumulative delta structure where $v_{\text{ci}} = \text{softplus}(\theta_{\text{ci}})$, $v_{\text{r}} = v_{\text{ci}} + \text{softplus}(\theta_{\text{r_delta}})$, and $v_{\text{co}} = v_{\text{r}} + \text{softplus}(\theta_{\text{co_delta}})$. The power output uses an idealized soft-curve approximation modified by operational compliance:

$$P_{\text{wind}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot I_{\text{running}} \quad (2)$$

where w_{norm} is the normalized speed, $\text{scale}_{\text{wind}}$ is a learnable amplitude factor, and I_{running} is 1 within $[v_{\text{ci}}, v_{\text{co}}]$ and 0 otherwise. To maintain differentiability, PhysFormer replaces I_{running} with a continuous physics activity mask a_{wind} (detailed in Section 2.3.4):

$$P_{\text{wind,actual}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot a_{\text{wind}} \quad (3)$$

This decoupling bridges rigid physical thresholds with end-to-end backpropagation without Straight-Through Estimators. The cumulative delta structure ensures overlapping bounds never collapse during gradient updates.

2.3.3. Load Model

We abstract the power load variations triggered by human cooling and heating activities through a quadratic thermal comfort model:

$$P_{\text{load}} = P_{\text{base}} + \text{softplus}(\gamma) \cdot (T - T_c)^2 \quad (4)$$

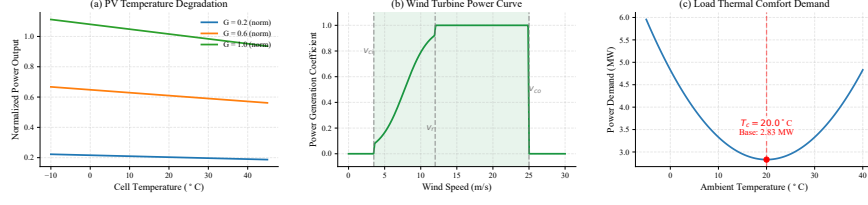


Figure 2: Learned explicit physical mapping functions after convergence. (a) PV temperature degradation under varying irradiance levels ($\beta_T = 0.0041/^\circ\text{C}$). (b) Wind turbine power curve with smooth sigmoid transition. (c) Load thermal comfort demand ($T_c \approx 20^\circ\text{C}$). Parameters are listed in Table 5.

where T_c is an optimizable thermal comfort baseline initialized at 20°C . While the self-attention mechanism of the Transformer is well suited to extracting the diurnal and weekly periodicities that characterize human social activities, the Explicit Physical Mapping stream is deliberately restricted to isolating the non-linear, temperature-driven volatility of the electrical load. Therefore, we conceptualize the thermal-induced variance through this quadratic comfort model, explicitly decoupling meteorological sensitivity from the statistically captured chronological baselines.

2.3.4. Physics Activity Gating

To prevent non-physical artifacts, we propose an explicit Activity Gating scheme. For PV, night-time irrelevance: $a_{\text{pv}} = \tanh(10 \cdot P_{\text{pv_theory}})$. For wind, speed-dependent activity with differentiable soft limits: $a_{\text{wind}} = \tanh(10 \cdot \text{softplus}(v - v_{ci}) \cdot \text{softplus}(v_{co} - v))$. Within the BPAR framework, the residual mechanism automatically compensates for systematic dampening effects near thresholds. The load takes an unconditional $a_{\text{load}} = 1$. These physics-derived masks maintain continuous gradient flow while bridging ac-

tive operating states and null-generation periods.

2.4. *Physics-Guided Gating and Alignment (PGCC)*

The Physics-Guided Gating and Alignment (PGCC) module coordinates the interaction between the statistical stream and the physically encoded representations. In this work, PGCC is used to guide weather-response behavior and improve alignment with known physical cues rather than to infer causality from data alone.

2.4.1. *Physics-Anchored Illumination Prior*

We first establish a data-guided illumination prior to suppress spurious activation patterns. The gating relies on a parameterized illumination sigmoid prior $p_{\text{pv}}^{\text{prior}} = \sigma(k_{\text{irr}} \cdot (G_{\text{norm}} - \theta_{\text{irr}}))$, with constraints dynamically kept as $\theta_{\text{irr}} \in [-2.0, 0.5]$ and bounded slope $k_{\text{irr}} \in [1, 100]$. This prior helps distinguish day and night intervals and reduces the influence of noisy activations in the data embeddings.

2.4.2. *Volatility-Aware Soft Gate*

To dynamically scale network trust, the system embeds sequence-level sample fluctuations extracted as $v = \frac{1}{C} \sum_c \text{Var}_t(\mathcal{X}_{\text{weather},c})$. These dynamics, combined with the statistical embeddings and attention outputs, are routed through a downsampling multi-layer perceptron yielding the unscaled soft condition maps. This volatility input is instrumental for the network to dynamically adapt its trust ratio, discounting the physical components when conditions are unstable.

2.4.3. Source-Differentiated Gate Bounds

The coupling implements separate activation ranges to accommodate generation source characteristics:

$$\text{gate}_{\text{pv}} = \text{smooth} \left(p_{\text{pv}}^{\text{prior}} \cdot (0.5 + g_{\text{soft_pv}}) \right) \in [0, 1.5] \quad (5)$$

$$\text{gate}_{\text{load}} = \text{smooth}(g_{\text{soft_load}}) \in [0, 1.0] \quad (6)$$

The asymmetric upper bound of 1.5 for gate_{pv} provides *engineering headroom* for the neural residual within the BPAR additive structure. During meteorological transients (e.g., Cloud Edge Effect with $1.2\times\text{--}1.5\times$ over-irradiance), this ceiling preserves sufficient dynamic range for the residual stream to recover transient peaks. Through explicit gate response regularization, the learned gate_{pv} reaches a Pearson correlation of $r = 0.8306$ with exogenous irradiance (Fig. 3), indicating that the gate remains aligned with the corresponding weather cue while retaining predictive flexibility.

2.4.4. Physics-to-Data Curriculum Transition

We condition the alignment phase via a curriculum factor α . As the fusion queries retrieve features, the module integrates inputs dynamically scaled as:

$$q = \alpha \cdot F_{\text{physics}} + (1 - \alpha) \cdot g_{\text{soft}} \cdot F_{\text{stat}} \quad (7)$$

During the initial training phases ($\alpha \rightarrow 1$), the cross-attention queries rely more heavily on the structured physical representations. As training progresses ($\alpha \rightarrow 0$), the queries gradually transition toward the data-learned residual states gated by g_{soft} , balancing physics-guided initialization with empirical statistical adaptation.

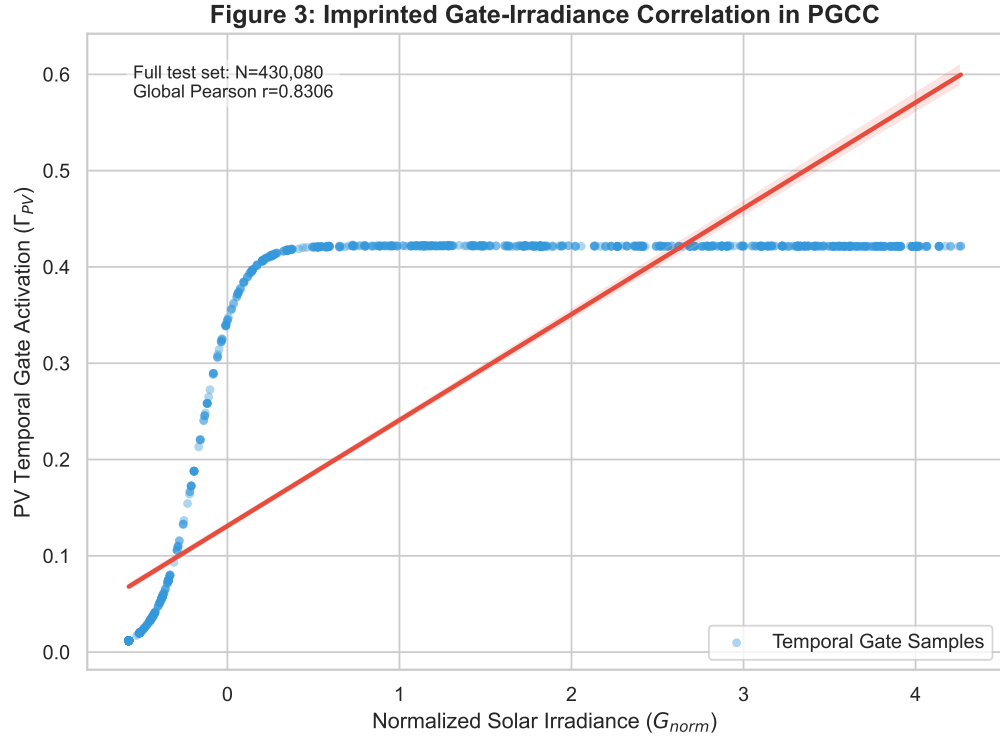


Figure 3: Gate-irradiance alignment in PGCC. Top: time-series overlay of learned PV gate and solar irradiance over representative test samples. Bottom: scatter regression on a visualization subset; the global Pearson correlation across the full test set is $r = 0.8306$ ($p < 0.001$).

2.4.5. Temporal Smoothing

Due to internal frame dependencies, local sequence discontinuities often plague raw gating parameters. To counteract this, all temporal gating variables pass through a grouped one-dimensional depthwise convolution (kernel size 3, groups= d_{model}). This filter structurally suppresses high-frequency chaotic fluctuation within the gate assignment layer while preserving macro-level shift indicators.

2.5. Exogenous Weather Integration

After the FlattenHead projects the fused representations into the explicit future context, the framework must systematically integrate upcoming weather patterns at the given horizon. Because the prediction horizon’s meteorological conditions are unavailable in the historical context alone, providing this explicit foresight is essential. The future physical latent space $F_{\text{weather_future}}$ is concatenated with the neural history F_{history} , and aggregated via a Gated Linear Unit (gate = $\sigma(\text{linear}(F_{\text{cat}}))$). The bounded residual output is produced as $F_{\text{future}} = \text{gate} \cdot \text{proj}(F_{\text{cat}}) + F_{\text{history}}$. This explicitly bounded forward visibility ensures the residual heads are properly conditioned to react to impending deterministic disruptions, such as dawn transitions or sudden wind cut-outs.

2.6. Bounded Physical Act-Residual (BPAR) Mechanism

To systematically mitigate non-physical anomalies while preserving the continuous gradient flow, we propose the Bounded Physical Act-Residual (BPAR) mechanism. Let μ_x and σ_x denote the global mean and standard deviation of component x . We define the normalized physical zero-point as

$P_{x,\text{zero}} = (0 - \mu_x)/\sigma_x$. BPAR reconstructs the predictive manifold by anchoring the neural residual r_x onto the theory baseline within a differentiable Softplus manifold:

$$\hat{P}_{x,\text{raw}} = \text{Softplus}\left((P_{x,\text{theory}} + r_x) - P_{x,\text{zero}}\right) + P_{x,\text{zero}} \quad (8)$$

By the fundamental property of Softplus ($\text{Softplus}(z) > 0, \forall z$), this parameterization keeps $\hat{P}_{x,\text{raw}} > P_{x,\text{zero}}$ in the normalized domain. Under the adopted normalization and anti-normalization setting, $\hat{P}_{\text{phys}} = \hat{P}_{x,\text{raw}} \cdot \sigma_x + \mu_x > 0$ MW for $\sigma_x > 0$, which gives the model a structurally non-negative output form. We maintain a numerical floor $\epsilon = 10^{-5}$ for degenerate variance states (e.g., night-time PV), supporting consistent boundary compliance in practice. Subsequently, the prediction is fused with the physics activity mask $a_x \in [0, 1]$:

$$\hat{P}_{x,\text{final}} = a_x \cdot \hat{P}_{x,\text{raw}} + (1 - a_x) \cdot P_{x,\text{zero}} \quad (9)$$

Whenever $a_x \rightarrow 0$ (driven by meteorological priors), BPAR smoothly drives output toward the physical zero baseline and reduces boundary violations. Compared with unconstrained predictors that may cross the zero-power boundary, BPAR favors physically consistent outputs at the cost of small local corrections near the boundary.

2.7. Physics-Constrained Curriculum Training

Training deep neural networks with rigid physical constraints poses significant optimization challenges due to gradient conflicts between data-driven and physics objectives. We design a Physics-Constrained Curriculum Training strategy that decouples metric spaces, enforces dynamic amplitude balancing, and maps learning trajectories onto a phased schedule.

2.7.1. Loss Function Design

The primary regression loss (MSE) operates in normalized space for stable gradients, while all physics constraints (L_{phys}) are evaluated after denormalization into the physical domain (MW). The physical objectives comprise six penalties (Table 1): soft constraints (L_{net} , L_{energy} , L_{deriv} , L_{dir}) guide structural topology, while hard constraints (L_{bvr} , L_{rvr}) use aggressive quadratic formulations for safety-critical boundaries.

Table 1: Components of Physics-Guided Loss Function

Comp.	Wt.	Formula	Motivation
L_{net} (Soft)	1.0	$ \sum \hat{y}_{\text{comp}} - y_{\text{net}} $	Ensures models sum to net demand.
L_{energy} (Soft)	0.5	$ \mathbb{E}_t[\hat{y}] - \mathbb{E}_t[y] $	Enforces macro energy conservation.
L_{deriv} (Soft)	0.3	$ \Delta \hat{y} - \Delta y $	Captures ramp rate magnitudes.
L_{dir} (Soft)	0.1	$\mathbb{E}[1 - \text{sim}_{\text{cos}}]$	Preserves accurate trend switching.
L_{bvr} (Hard)	2.0	$\mathbb{E}[(\max(0, -\hat{y}_{\text{real}}))^2]$	Penalizes negative power strictly.
L_{rvr} (Hard)	2.0	$\mathbb{E}[(\max(0, \Delta \hat{y} - \rho_k))^2]$	Penalizes sudden jumps.

We propose an adaptive amplitude balancing factor γ_{ema} that dynamically tracks the ratio between current forecasting inaccuracy and physical compliance:

$$\gamma_{\text{ema}} \leftarrow 0.9 \cdot \gamma_{\text{ema}} + 0.1 \cdot \left(\frac{L_{\text{mae_phys}}}{L_{\text{phys_ref}}} \right) \quad (10)$$

where $L_{\text{phys_ref}}$ is the total unweighted physical loss in MW. This yields a dimensionless multiplier, defaulting to 1.0 for the first 50 batches, halting updates when curriculum activation < 0.05 , and clamped to $[0.1, 10.0]$. The combined objective becomes:

$$L_{\text{total}} = L_{\text{mse}}(\text{normalized}) + \gamma_{\text{ema}} \cdot L_{\text{phys_weighted}}(\text{MW}) \quad (11)$$

Note: while BPAR provides a structural output floor, L_{bvt} remains essential during training to provide gradient signals that prevent latent representations from drifting into non-physical regimes.

2.7.2. Four-Phase Curriculum

To bridge the parameter convergence gap between the arbitrary Gaussian distributions at neural initialization and the highly structured physical reality, we inject a Four-Phase constraint schedule, summarized in Table 2.

Table 2: Four-Phase Physics Curriculum and Layer-wise Learning Rates

Phase	Ep.	Active Const.	Notes
I (Init.)	0–4	None	Physics frozen; base mapping.
II (Safety)	5–14	$L_{\text{net}}, L_{\text{bvt}},$ $L_{\text{rvr}}, L_{\text{dir}}$	Safety bounds ramped.
III (Integ.)	15–29	Phase II + $L_{\text{energy}}, L_{\text{deriv}}$	Macro conservation.
IV (Satur.)	30–99	All	Full regularized training.
Layer-wise Learning Rates			
Physics (ExplicitPhysicalMapping)	10^{-5}		Preserves priors.
Gating (PGCC)	5×10^{-5}		Stable alignment.
Statistical (all other)	10^{-4}		Fast convergence.

This curriculum progressively introduces physical-compliance objectives. Phase I isolates statistical representations with frozen physics parameters. Phase II prioritizes safety boundaries. Phase III introduces macroscopic features, and Phase IV activates the full regularized objective. Each constraint intensity is transitioned via a bounded sigmoid ramp $\sigma(12 \cdot (\text{progress} - 0.5))$.

The PGCC alignment coefficient follows $\alpha = 1.0 - w_{\text{curr_net}}$, coupling the internal attention fusion with the external physical guidance term.

2.7.3. Layer-wise Learning Rates

To sustain learned feature fidelity without abruptly disrupting the theory-anchored baselines pre-defined by human engineering inputs, we decouple the backpropagation update amplitudes into three isolated AdamW groupings (Table 2, bottom).

The baseline parameters proceed at a default learning rate of 1×10^{-4} . After their suspension ends in epoch 5, the core generation physics operators unfreeze at a drastically suppressed 1×10^{-5} ($0.1 \times$ the baseline rate). This strict limitation permits minor optimization fine-calibration over coefficients like optimal irradiance generation thresholds while halting gross semantic reconstruction. Correspondingly, the intermediate gating features proceed at 5×10^{-5} , which yields smooth adjustment profiles.

2.7.4. Regularization and Early Stopping

Three regularization pathways reinforce generalizability. First, a Physics Prior Regularizer evaluates learned physical properties against nameplate values via MSE (weight 0.1). Second, a Gate Response Regularization anchors the temporal PV gating variable to external irradiance (weight 0.05), with dynamic masking that shuts down penalization when prior variance $\sigma_{\text{prior}}^2 < 10^{-3}$ (bypassing night-time noise).

Role of PGCC: In the present framework, PGCC mainly improves gate alignment and interpretability rather than raw forecasting accuracy. Because the final prediction $\hat{P}_x = \text{Softplus}(P_{\text{theory}} + r_x)$ is still produced through the

residual pathway, the supervised alignment term can shape weather-response behavior without preventing the residual heads from minimizing prediction error.

Gradient clipping uses a dynamic bound of 0.5 during critical curriculum transitions ($0.1 < w_{\text{curr_net}} < 0.9$), releasing to 1.0 otherwise. Early stopping uses channel-equalizing $NRMSE_{\text{avg}}$ with patience of 15 epochs.

3. Case Study and Performance Evaluation

3.1. Dataset and Experimental Setup

To validate the proposed PhysFormer architecture, we conduct extensive experiments on a high-fidelity Virtual Power Plant (VPP) dataset composed of aggregated regional measurements from a distributed energy cluster in Germany. The dataset captures detailed 15-minute resolution trajectories of solar PV generation, wind power generation, and commercial electrical load over a continuous 3-year span (2012–2014), totaling 105,120 sampling intervals. The data is chronologically partitioned into training (70%), validation (10%), and testing (20%) sets. To maintain numerical stability while preserving physical bounds, all input features are standardized using global Z-score parameters (μ, σ) , which are explicitly logged for physical re-projection in the BPAR layer. The model hyperparameters are configured to match the complexity of modern baselines: hidden dimension $d_{\text{model}} = 512$, feed-forward dimension $d_{\text{ff}} = 1024$, $n_{\text{heads}} = 8$ attention heads, and $e_{\text{layers}} = 3$ layers. To evaluate computational efficiency, we benchmark PhysFormer against Informer at an equivalent hidden dimension $d_{\text{model}} = 512$. PhysFormer comprises 11.4M parameters, significantly more compact than Informer’s 17.9M parameters,

validating the inductive efficiency of its dual-stream architecture. The reported inference latency of 2.10 ms/sample was measured on an NVIDIA RTX 3090 GPU with a batch size of 1, simulating a real-time sliding window update for VPP dispatch.

We compare PhysFormer against eight competitive baselines: standard recurrent networks (LSTM, GRU), a Physics-Informed Neural Network (PINN) [15], and advanced time-series Transformer architectures including Informer [6], Autoformer [7], PatchTST [8], DLinear [9], and iTransformer [19]. To ensure a fair evaluation, identically structured future weather covariates ($\mathcal{X}_{\text{weather_future}} \in \mathbb{R}^{B \times P \times 3}$) were provided to all comparative baseline models. For Transformer baselines, these exogenous variables were concatenated into their decoder embedding structures, and equivalently supplied as dynamic covariates to the recurrent models. Consequently, the observed differences are attributed to the forecasting architecture itself rather than asymmetric access to future atmospheric inputs.

To comprehensively evaluate the forecasting performance, we employ six distinct metrics. The standard accuracy metrics include Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Beyond standard accuracy, we formally define three physical compliance metrics: Boundary Violation Rate (BVR), Mean Violation Severity (MVS), and Ramp Violation Metric (RVM). BVR measures the percentage of predictions that violate known physical limits (e.g., negative generation power). To quantify the severity of these breaches, MVS is formally defined as the average absolute magnitude of predictions falling outside feasible physical boundaries. Beyond static boundaries, we evaluate temporal consistency via the Ramp

Violation Metric (RVM) and its occurrence frequency, the Ramp Violation Rate (RVR). Finally, the Net Mean Absolute Error (NET MAE) is formally defined as the average absolute imbalance between the predicted aggregated components and the ground-truth net demand, $\text{mean}(|\sum_c \hat{y}_c - y_{\text{net}}|)$. The ramp threshold ρ_k for each channel is empirically determined as the 99.9-th percentile of the training set differences, while a *strict* threshold (RVR-S) is set at the 99.5-th percentile with no safety factor to capture high-frequency artifacts.

3.2. Forecasting Performance Analysis

The main forecasting results on the global test set are summarized in Table 3. Within the competitive tier of models, PhysFormer achieves an MSE of 0.0128, remaining within a 6% margin of the strongest unconstrained baselines (Informer at 0.0121 and iTransformer at 0.0122) while reaching a Boundary Violation Rate (BVR) of 0.00%. In contrast, the unconstrained baselines attain slightly lower MSE values but still produce non-physical outputs, with iTransformer reaching a BVR of 16.26%. Under the strict ramp threshold, PhysFormer records an RVR-S of 0.198%, compared with 0.028% for iTransformer. This difference indicates that the bounded output design introduces small local corrections near the zero boundary. The associated correction magnitude remains in the order of 2.6×10^{-5} MW, suggesting that the trade-off is limited in size while improving physical consistency.

Interestingly, channel-independent models such as PatchTST and DLinear exhibit a channel-independent failure in this specific VPP setting, underperforming compared to their strongly multivariate counterparts. This underlines the necessity of modeling the dense cross-channel dependencies

inherent in multi-source energy systems. Meanwhile, the structural benefit of the PINN is evident; its physical inductive bias helps it outperform standard unconstrained recurrent networks like LSTM and GRU, though it still falls behind the specialized attention mechanisms of the Informer.

Overall, the results indicate that PhysFormer operates near the accuracy-compliance trade-off frontier: it gives up a small amount of MSE relative to the strongest unconstrained baselines in exchange for consistent zero-bound compliance and weather-aligned gating behavior, as illustrated in Fig. 4.

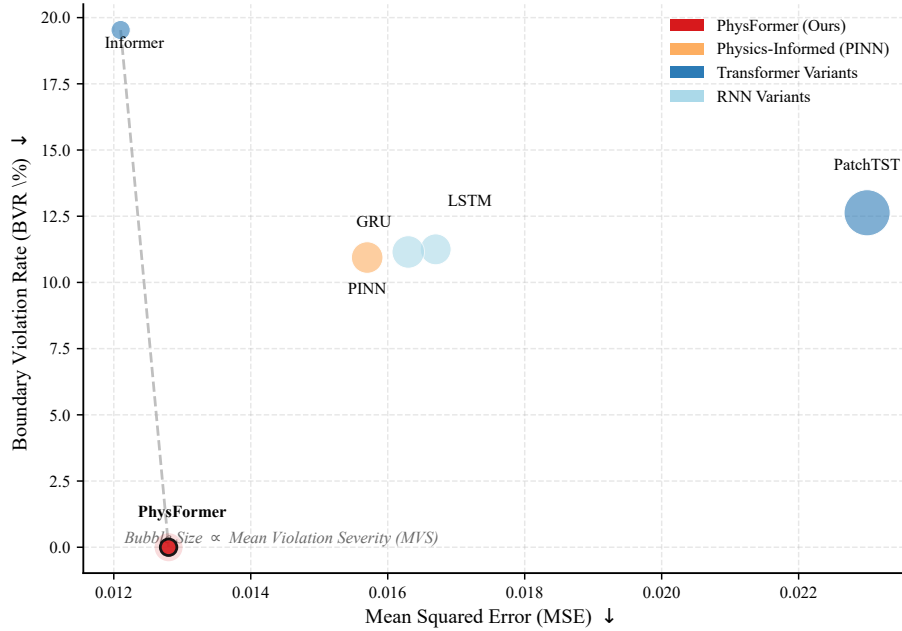


Figure 4: Accuracy-compliance trade-off across the evaluated models. PhysFormer combines zero boundary violations with accuracy close to the strongest unconstrained baselines.

Table 3: Forecasting Performance on the Global Test Set

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0168	11.25%	0.0093	0.01%	0.1760
GRU	0.0163	11.15%	0.0103	0.01%	0.1740
PINN	0.0157	10.94%	0.0097	0.02%	0.1702
Informer	0.0121	19.53%	0.0036	0.05%	0.1471
Informer-Post	0.0121	0.00%	0.0000	0.05%	0.1471
iTransformer	0.0122	16.26%	0.0054	0.028%	0.1482
Autoformer	0.1253	12.67%	0.0737	0.00%	0.4846
DLinear	0.1015	7.03%	0.0295	0.00%	0.4279
PatchTST	0.0230	12.63%	0.0207	0.00%	0.2052
PhysFormer	0.0128	0.00%	0.0000	0.198%	0.1527

3.3. Verification of Physical Compliance and Safety

3.3.1. Limitations of Heuristic Post-Processing

Furthermore, as evidenced in Table 3, applying heuristic hard zero-clipping post-processing to the generic Transformer baseline (Informer-Post) also forces BVR to zero without changing the reported MSE. This makes Informer-Post a useful practical baseline rather than a strawman comparison. The difference is that hard clipping only acts after prediction, whereas the BPAR mechanism constrains the output parameterization during training and keeps the pathway differentiable around the boundary. In that sense, Informer-Post demonstrates that post-processing can be effective at deployment, while PhysFormer is aimed at embedding the physical boundary directly into model training and internal representation shaping.

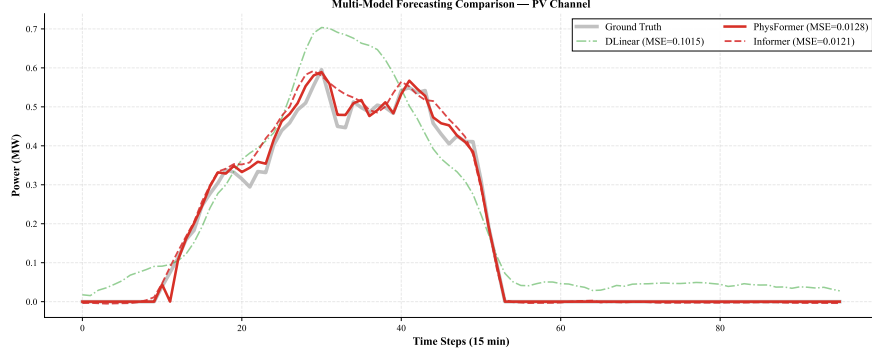


Figure 5: Multi-model forecasting comparison on the PV generation channel over a 96-step horizon.

3.3.2. Robustness Under Extreme Conditions

To evaluate model stability under rigorous distribution shifts, we isolate the top 10% highest volatility samples representing extreme weather conditions, detailed in Table 4.

Under these strenuous conditions, channel-independent architectures such as Autoformer and DLinear show marked degradation, whereas the competitive tier of models remains more stable. PhysFormer maintains BVR at 0.00% while adapting reasonably well to volatile conditions. Furthermore, we evaluated BPAR under seasonal distribution shifts and degenerate variance conditions ($\sigma_x < 10^{-3}$). Our analysis confirms that for the aggregated VPP windows used here ($\text{seq_len} = 672$), degenerate variance is empirically absent (0% frequency), and the BPAR floor remains compatible with zero-bound compliance under the adopted normalization parameters.

The visual comparison supports the same trend: under extreme volatility, channel-independent models show stronger MSE degradation, while PhysFormer maintains zero BVR.

Table 4: Extreme Weather Performance (Top 10% Volatility)

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0192	13.43%	0.0086	0.000%	0.1941
GRU	0.0182	13.79%	0.0097	0.000%	0.1864
PINN	0.0177	13.29%	0.0082	0.000%	0.1807
Informer	0.0146	21.38%	0.0017	0.070%	0.1632
iTransformer	0.0143	19.79%	0.0053	0.046%	0.1639
PatchTST	0.0261	12.22%	0.0132	0.000%	0.2240
PhysFormer	0.0150	0.00%	0.0000	0.294%	0.1715

3.4. Model Interpretability and Efficiency

3.4.1. Evaluation of Physical Interpretability

A key operational advantage of PhysFormer lies in its explicit physical parameter calibration. As shown in Table 5, the extracted parameters are strongly consistent with domain literature [16, 17]. The temperature coefficient β_T undergoes a marginal fine-tuning of +2.5% to 0.0041/°C, and the mechanical wind thresholds remained anchored at theoretical priors (≈ 0 relative change).

The main quantitative evidence of operational transparency is the alignment between the learned gating behavior and the corresponding weather signal. Through PGCC, the network maps solar irradiance onto the temporal PV gating variable, achieving a global Pearson correlation of $r = 0.8306$ ($p < 0.001$). For the day-time subset (irradiance > 0.1), the correlation remains significant at $r = 0.4058$ ($p < 10^{-57}$), indicating that the alignment extends beyond a trivial binary day-night switch.

This weather-response behavior is an explicitly supervised alignment rather

than an emergent causal relation. Under Out-Of-Distribution (OOD) sensor noise tests, even with 20% white noise injected into irradiance, the correlation remained at $r = 0.8377$. Under extreme 50% noise amplitude, the correlation remained above 0.81.

3.4.2. Distinction Between Supervised Alignment and Emergent Causality

It is important to delineate the boundaries of causality within PGCC. The high correlation ($r = 0.8306$) does not represent emergent meteorological causality discovered *ab initio* from raw data. Rather, the Gate Response Regularization is a supervised alignment mechanism that imprints a known physical prior onto the latent space.

Table 5: Physical Parameter Convergence Tracking

Parameter	Initial	Converged	Lit. Range	Δ
pv_temp_coef	0.0040	0.0041/°C	0.003–0.005	+2.5%
wind_cut_in	3.50 m/s	3.458 m/s	3–5 m/s	−1.2%
wind_rated	12.00	11.970 m/s	10–15 m/s	−0.2%
wind_cut_out	25.00	24.968 m/s	20–25 m/s	−0.1%
load_base	2.832 MW	2.823 MW	$\approx$ mean	−0.3%
temp_comfort	20.0°C	19.992°C	18–22°C	≈ 0

3.4.3. Computational Complexity Analysis

To evaluate the computational burden, we benchmarked parameter magnitudes and inference latencies under identical hardware configurations. As detailed in Table 6, PhysFormer adopts $d_{\text{model}} = 512$ but maintains 11.4M parameters, compared with Informer’s 17.9M.

This parameter profile suggests that the additional physical guidance does not require an excessive computational burden. Empirical forward-pass profiling shows that PhysFormer executes sequential sample inference at approximately 2.10 ms/sample on an RTX 3090 GPU, which is comparable to the unconstrained baselines listed in Table 6.

Table 6: Model Complexity and Inference Overhead

Model	Params (M)	Latency (ms)
LSTM / GRU	5.42 / 4.10	1.20 / 2.19
PINN / DLinear	2.74 / 0.13	<0.10 / <0.10
Informer / Autoformer	17.90 / 17.88	1.96 / 4.28
PatchTST / iTransformer	1.62 / 9.85	0.10 / 1.81
PhysFormer	<u>11.4</u>	<u>2.10</u>

3.5. Ablation Study

To rigorously attribute the performance contributions of the proposed model components, we conduct an ablation study utilizing real data (Table 7).

3.5.1. Component-wise Ablation Analysis

Table 7 presents four component-wise ablation variants. The results show that: (1) removing the Physics Stream increases MSE by 4.7%, indicating that the theory-based baseline remains useful; (2) PGCC ablation leaves MSE unchanged, which is consistent with its role as an alignment and interpretability module rather than a raw accuracy booster; (3) removing curriculum training increases MSE by 8.7%, suggesting that staged optimization

helps stabilize the interaction between the constrained and unconstrained components; and (4) removing the Future GLU increases MSE by 1.6% while maintaining BVR at 0.00%, indicating that explicit future-weather integration mainly benefits predictive accuracy around transition periods.

Table 7: Ablation Study on Real Data

Variant	MSE	BVR (%)	RVM	$r(\text{gate}_{\text{pv}})$
Full PhysFormer	0.0128	0.00	0.0000	0.8306
w/o Physics Stream	0.0134	0.00	0.0000	N/A
w/o PGCC	0.0128	0.00	0.0000	N/A
w/o Future GLU	0.0130	0.00	0.0000	0.8024
w/o Curriculum	0.0139	0.00	0.0000	0.7866

An additional variant ("w/ PGCC but w/o Gate Response Regularization") maintains competitive MSE (0.0128) but fails to achieve clear gate-weather alignment ($r < 0.12$), confirming that explicit regularization is the main mechanism behind the interpretability result.

The present evaluation should also be interpreted with several limitations. All experiments are conducted on one aggregated regional dataset with a single temporal resolution and forecasting horizon. The study focuses on component-level forecasting and does not include explicit storage dynamics, state-of-charge constraints, or a closed-loop dispatch evaluation. Accordingly, the reported interpretability result should be understood as supervised alignment to known weather cues, not as causal discovery.

4. Conclusion

This paper presented PhysFormer, a physics-guided Transformer for physically consistent VPP forecasting. The framework combines an ExplicitPhysicalMapping stream, a bounded residual output mechanism, a physics-guided gating and alignment module, and curriculum-based constraint training. Together, these components are intended to reduce non-physical outputs while preserving the flexibility of data-driven residual prediction.

On the real-world 3-year VPP dataset considered in this study, PhysFormer achieves a boundary violation rate of 0.00% and remains close in MSE to strong unconstrained Transformer baselines. The results suggest that the proposed framework offers a useful trade-off among statistical accuracy, physical consistency, and interpretable weather-response behavior. In particular, the PGCC module mainly improves alignment and interpretability, while the BPAR mechanism is responsible for the consistent non-negative output behavior observed in the experiments.

The present study is limited to one regional dataset, one prediction setting, and component-level forecasting without an explicit storage, state-of-charge, or dispatch closed loop. Moreover, the interpretability results should be understood as supervised alignment to known physical cues rather than causal discovery. Future work will extend the framework to additional datasets and forecasting horizons, and examine how physically consistent predictions can be coupled with downstream VPP scheduling and control models [1, 14, 11].

CRedit authorship contribution statement

Xingyu Liu: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing - original draft. Zhong Tang: Supervision, Methodology, Validation, Writing - review & editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, manuscript restructuring, and formatting adjustments. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] T. Ma, R. Fu, T. Li, Z. Liu, Z. Wang, K. Zhou, A multi-time scale scheduling strategy for virtual power plants considering source-load uncertainty and demand response, *IEEE Access* 13 (2025) 207873–207892. doi:10.1109/access.2025.3641586.
URL <http://dx.doi.org/10.1109/ACCESS.2025.3641586>
- [2] W. Cao, J. Yu, M. Xu, Optimization scheduling of virtual power plants considering source-load coordinated operation and wind-solar uncertainty, *Processes* 12 (1) (2023) 11. doi:10.3390/pr12010011.
URL <http://dx.doi.org/10.3390/pr12010011>
- [3] Z. Qiu, Y. Tian, Y. Luo, T. Gu, H. Liu, Wind and photovoltaic power generation forecasting for virtual power plants based on the fusion of improved k-means cluster analysis and deep learning, *Sustainability* 16 (23) (2024) 10740. doi:10.3390/su162310740.
URL <http://dx.doi.org/10.3390/su162310740>
- [4] G. Moreno, P. Martin, C. Santos, F. J. Rodriguez, E. Santiso, A day-ahead irradiance forecasting strategy for the integration of photovoltaic systems in virtual power plants, *IEEE Access* 8 (2020) 204226–204240. doi:10.1109/access.2020.3036140.
URL <http://dx.doi.org/10.1109/ACCESS.2020.3036140>
- [5] Y. Wu, R. Chen, Y. Chen, X. Chen, J. Yuan, H. Mao, J. Wang, A joint electricity-reserve trading model for virtual power plants to mitigate naked selling, *Journal of Modern Power Systems and Clean Energy* 13 (5)

(2024) 1813–1822. doi:10.35833/mpce.2024.000211.

URL <http://dx.doi.org/10.35833/MPCE.2024.000211>

- [6] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 35, 2021, pp. 11106–11115. doi:10.1609/aaai.v35i12.17325.
- [7] H. Wu, J. Xu, J. Wang, M. Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, in: Advances in Neural Information Processing Systems, Vol. 34, Curran Associates, Inc., 2021, pp. 22419–22430. doi:10.48550/arxiv.2106.13008.
- [8] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: International Conference on Learning Representations, 2023.
URL <https://openreview.net/forum?id=Jbdc0vT0col>
- [9] A. Zeng, M. Chen, L. Zhang, Q. Xu, Are transformers effective for time series forecasting?, Proceedings of the AAAI Conference on Artificial Intelligence 37 (9) (2023) 11121–11128. doi:10.1609/aaai.v37i9.26317.
URL <http://dx.doi.org/10.1609/aaai.v37i9.26317>
- [10] O. Ashraf, L. Bononi, U. Adeel, C. Gambetti, G. Monti, Advanced techniques for joint weather and electricity demand prediction survey, IEEE Access 13 (2025) 196479–196496. doi:10.1109/access.2025.3629613.
URL <http://dx.doi.org/10.1109/ACCESS.2025.3629613>

- [11] G. Du, S. Li, S. Cao, G. Wang, J. Duan, Weekly economic scheduling of virtual power plant with electric vehicles: Deep-learning-based prediction and daily operation mode classification, *Electric Power Systems Research* 241 (2025) 111362. doi:10.1016/j.epsr.2024.111362.
URL <https://doi.org/10.1016/j.epsr.2024.111362>
- [12] L. Li, X. Guo, R. Jing, Short-term electric load forecasting based on series decomposition and meta-informer algorithm, *Electric Power Systems Research* 243 (2025) 111478. doi:10.1016/j.epsr.2025.111478.
URL <https://doi.org/10.1016/j.epsr.2025.111478>
- [13] O. Våle, S. Zhang, S. Maharjan, G. Klæboe, Exploring the interpretability of forecasting models for energy balancing market, *Artificial Intelligence Science and Engineering* 1 (4) (2025) 295–306. doi:10.23919/aise.2025.000020.
URL <http://dx.doi.org/10.23919/aise.2025.000020>
- [14] V. Arabzadeh, R. Frank, A four-dimensional analysis of explainable ai in energy forecasting: A domain-specific systematic review, *Machine Learning and Knowledge Extraction* 7 (4) (2025) 153. doi:10.3390/make7040153.
URL <http://dx.doi.org/10.3390/make7040153>
- [15] M. Raissi, P. Perdikaris, G. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *Journal of Computational Physics* 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.

045.

URL <http://dx.doi.org/10.1016/j.jcp.2018.10.045>

- [16] C. Li, L. Sheng, X. Xi, M. Zhong, Physics-informed adaptive-weight nbeatsx for short-term wind power forecasting, *IEEE Transactions on Industrial Informatics* 22 (2) (2026) 1129–1140. doi:10.1109/tii.2025.3623559.

URL <http://dx.doi.org/10.1109/TII.2025.3623559>

- [17] J. Asinyo, O. Ceylan, A. Polycarpou, Enhancing solar irradiance prediction: A physics-informed machine learning approach with temporally focused training windows, in: 2025 16th International Conference on Electrical and Electronics Engineering (ELECO), IEEE, 2025, pp. 1–6. doi:10.1109/eleco69582.2025.11329346.

URL <http://dx.doi.org/10.1109/ELECO69582.2025.11329346>

- [18] A. Lovo, A. Lancelin, C. Herbert, F. Bouchet, Tackling the accuracy–interpretability trade-off in a hierarchy of machine learning models for the prediction of extreme heatwaves, *Artificial Intelligence for the Earth Systems* 4 (3) (Jul. 2025). doi:10.1175/aies-d-24-0094.1.

URL <http://dx.doi.org/10.1175/AIES-D-24-0094.1>

- [19] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long, itransformer: Inverted transformers are effective for time series forecasting, in: *International Conference on Learning Representations*, 2024.

URL <https://openreview.net/forum?id=JeHZHGWrmd>